

I'M THE FIRST STEP, THE GUIDING LIGHT, CHOOSE ME WRONG, AND NOTHING'S RIGHT. NARROW OR BROAD, I SET THE TONE—YOUR RESEARCH STARTS WITH ME ALONE. WHAT AM I?

Step-by-Step Guide to Selecting a Research Topic for Undergraduates

Empowering Students to Choose Purposeful and Practical Research Topics

A Guide for First-Year BDS Students

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🎯LEARNING OBJECTIVES

By the end of this session, students will be able to:

1. **Understand the Importance** of selecting a relevant and feasible research topic in their field of study.
2. **Identify Key Factors** to consider when choosing a research topic (e.g., interest, originality, resources, relevance).
3. **Explore Sources** for topic ideas such as literature, clinical observations, mentors, and recent trends.
4. **Apply a Step-by-Step Process** to narrow down and refine a broad area of interest into a specific, researchable topic.
5. **Evaluate the Feasibility** of a selected topic based on scope, time, access to data, and supervisor support.
6. **Formulate a Preliminary Research Question** or hypothesis based on the chosen topic.

Where Does Topic Selection Fit?

Overview of research process:

- **Topic Selection (Our Focus Today!)**
- Literature Review
- Formulating Research Question
- Study Design
- Data Collection
- Data Analysis
- Writing & Dissemination



Every great discovery starts with a curious mind and a well-chosen question."



A poor topic choice is like being lost at a crossroads — no direction leads you forward.”

Seeing Research Opportunities Everywhere

Observe clinical challenges and common patient complaints.

Daily life oral hygiene practices or habits may raise questions.

Listen to faculty and senior student discussions.

Read alot

Keeping It Real: What Can YOU Do?

Choose manageable topics.

Focus on surveys, records, simple observations.

Avoid complex clinical trials or expensive equipment.

Think about participants /data access.

Is Your Topic FINER?

FINER Criteria: Feasible, Interesting, Novel, Ethical, Relevant

F – Feasible

- Can you carry out the study with the time, resources, and skills available?
  (e.g., sample size, equipment, data access)

I – Interesting

- Does the topic spark curiosity—for **you** and **others**?
  (Avoid something that feels dull or forced.)

N – Novel

- Is it new or offering a fresh angle?
  (Not necessarily groundbreaking, but should add something unique to existing knowledge.)

E – Ethical

- Can it be done without harming participants or violating ethical standards?
  (Approval from ethical boards, informed consent, etc.)

R – Relevant

- Will it contribute to science, practice, policy, or future research?
  (It should matter in your field or community.)

The 'Good Topic' Checklist (PICO Concept)

The PICO concept is a helpful tool for framing a focused clinical research question, especially in evidence-based practice. Here's what it stands for:

P – Patient / Problem / Population

Who is the patient or population?

What is the condition or issue?
Example: Adults with high blood pressure

I – Intervention

What is the main intervention or treatment being considered?
Example: Daily exercise

C – Comparison

What is the alternative to compare with the intervention? (if any)
Example: No exercise or medication

O – Outcome

What result are you hoping to measure?
Example: Reduction in blood pressure

Research Ethics: Non-Negotiable

Consent and confidentiality are essential.

Do no harm.

IRB approval is required for human research.

Avoid exploitative or harmful topics.

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UNDERSTAND REQUIREMENT

BRAINSTORM BROAD AREA

NARROW DOWN THE AREA

QUICK LITERATURE REVIEW

CHECK FEASIBILITY

RESEARCH QUESTION

GET FEED BK

STEPS TO CHOOSE THE RIGHT TOPIC



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Step-by-Step Guide to Selecting a Research Topic
Step 1: Understand the Requirements

- **Know the scope**: Are there specific guidelines from your teacher/professor/advisor?
- **Know the type**: Is it theoretical, experimental, applied, or review-based?
- **Know the length and deadline**: How long do you have to complete it?

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Step 2: Brainstorm Broad Areas of Interest

Think about:

- Subjects you enjoy.
- Problems you've noticed in real life.
- Current trends in your field or news.

Examples of broad areas?

Endodontics

•Root canal disinfection techniques
•Instrumentation systems (rotary vs manual)
•Periapical lesion healing
•Pulp regeneration

Restorative Dentistry

•Composite vs amalgam restorations
•Microleakage in dental materials
•Adhesive systems
•Minimally invasive dentistry

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Step 3: Narrow Down the Area

Use these filters:

What's trending? (new tech, social issues, recent laws, etc.)

What's underexplored?

What aligns with your future goals or career?

For example: THINK

- From "Artificial Intelligence" → "Ethics in AI" or "AI in Education"
- From "Public Health" → "Mental Health in Teenagers Post-COVID"

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Step 4: Do a Quick Literature Review

Use **Google Scholar**, **PubMed**, **ScienceDirect**

Look for:

- Gaps in research
- Common methodologies
- Recent developments

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Step 5: Check Feasibility

Ask yourself:

- Do I have access to data, tools, or labs?
- Is it too broad or too narrow?
- Can I complete it within the deadline?

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Step 6: Draft a Preliminary Research Question

A good research question is:

- **Clear**: Not vague.
- **Focused**: Narrow enough to answer in detail.
- **Researchable**: Can be answered with data or literature.
- **Significant**: Contributes something to your field.

Example:
Instead of: “How does AI work?”
Try: “How can AI-powered chatbots improve mental health support for university students?”

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Step 7: Get Feedback

- Ask your advisor or peers for input.

Make sure it's academically valuable and doable

Your Turn: Discovering Your Research Spark!

Activity:

- Think of last 3 dental lectures/ practicals.

- Write down 1 question or concept from each.

- Add 1 oral health idea from daily life.

Thank You!

I wish you all the best in your research endeavors!

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